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Report Information

Prelim. Report Publish Date	9/2/2015
SI Number	SI38944
Analyst Name	Tim M Mate

Product Description

Product Type	Smartphone
Brand	Samsung
Device Name	Z1
Device Model #	SM-Z130H/DS
Official Release Date	1/14/2015
Target Market	India
Retail Price	\$76.75
Weight (grams)	112.4 (Measured)
Device Dimensions (mm)	120.5 x 63.1 x 10.0 (Measured at Longest/Widest/Thickest Points)

Product Features

Operating System	Tizen 2.3
Processor Spec	1.2 GHz Dual-Core Spreadtrum
RAM Support	0.75 GB
Communications	GSM: 850/900 LTE: 900/2100 DCS: 1800, PCS: 1900
Connectivity	WiFi 802.11b/g/n, Bluetooth 4.0 w/ A2DP, microUSB 2.0, FM Radio, GPS
User Interface	Capacitive Keys & Touchscreen
Storage	Internal: 4 GB ROM External: microSD (max. 64 GB)
Sensors	3-Axis MEMS Accelerometer, VGA Front Camera Sensor
Battery Type	3.8 V, 1500 mAh, Li-ion
Battery Life (Hrs.)	Use Time: 7; Standby Time: Unknown
Display	4" PLS TFT-LCD, 800 x 480 Pixels, 16,777,216 Colors
Rear Camera	3.1 MP CMOS, LED Flash, Video Capability

Preliminary Report

Discussion

Dear Client:

Preliminary reports are meant to help you quickly identify the major IC wins at a system level. Such ICs are usually found on the main board PCB or other system-level substrates. The ICs found within a subsystem, such as a camera or display subsystem, are not considered a major IC with regard to a device's overall system. The reasoning is that while the preliminary report tries to identify the major ICs unique to a device, subsystems may be used in several different devices. However, in the final report, all ICs and modules are identified, including those found within subsystems.

A major IC may be categorized as follows:

Component Function:

	Applications & Baseband Processor
	Application Processor
	Baseband Processor
	Camera / Image
	Connectivity
	Display / Touchscreen
	Logic
	Memory: Mixed
	Memory: Non-Volatile
	Memory: Volatile
	Mixed Signal
	Other
	Power Management / Audio
	RF Component
	Sensor

Please note that the information in this report should be considered preliminary. Although every effort has been made to ensure the accuracy of the data, this device is currently still in analysis. As the device is further analyzed, we reserve the right to update the data as needed without notification.

Once the analysis of our device is complete, we will publish a full report. In the event the information in the final report does not match the information found in this preliminary report, the final report and all of its content will supersede the information found in the preliminary report.

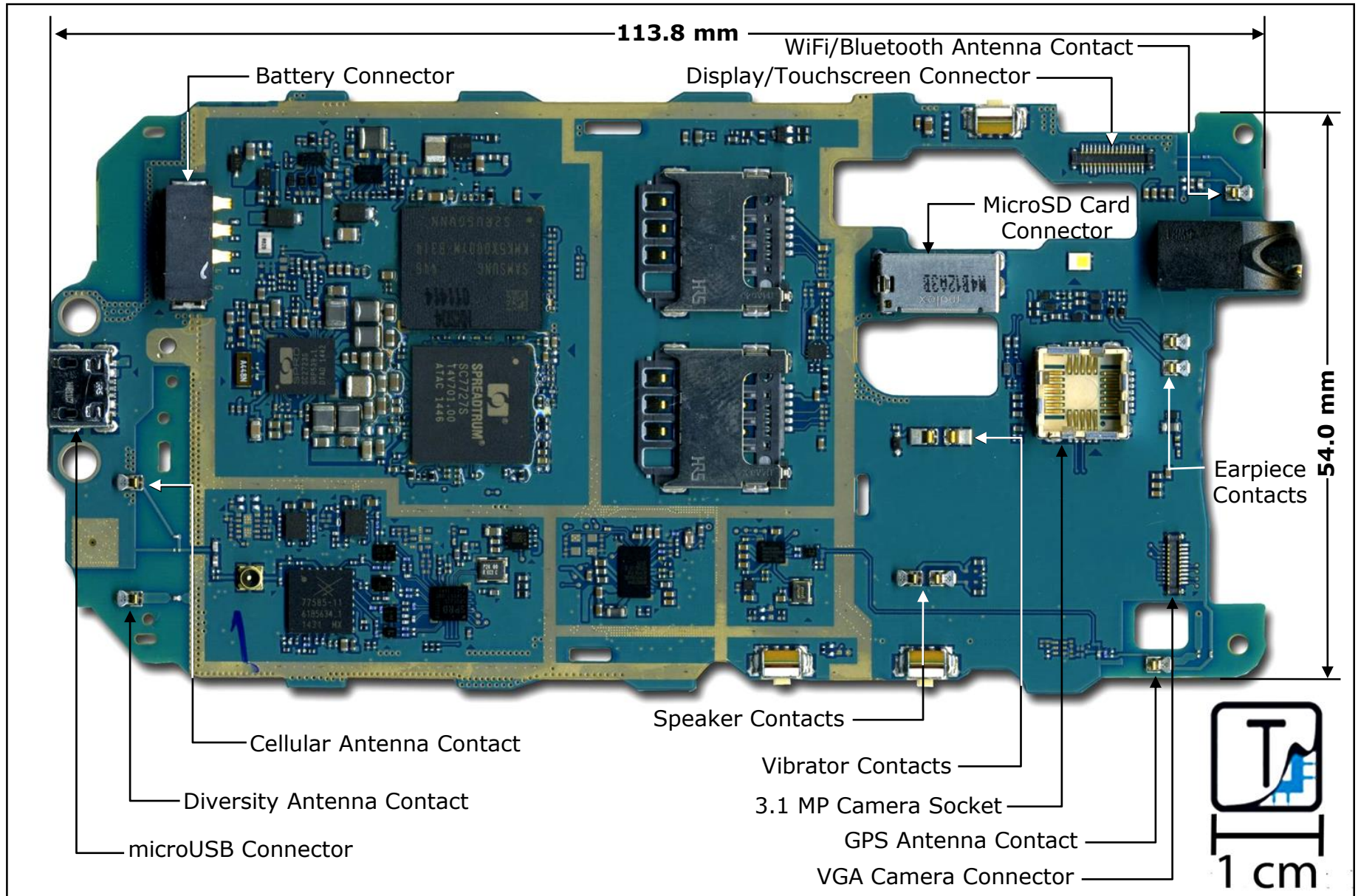
Identification by Component Function

In both the 2015 Deep Dive and Survey Plus reports, ICs and modules are categorized by their function within the system.

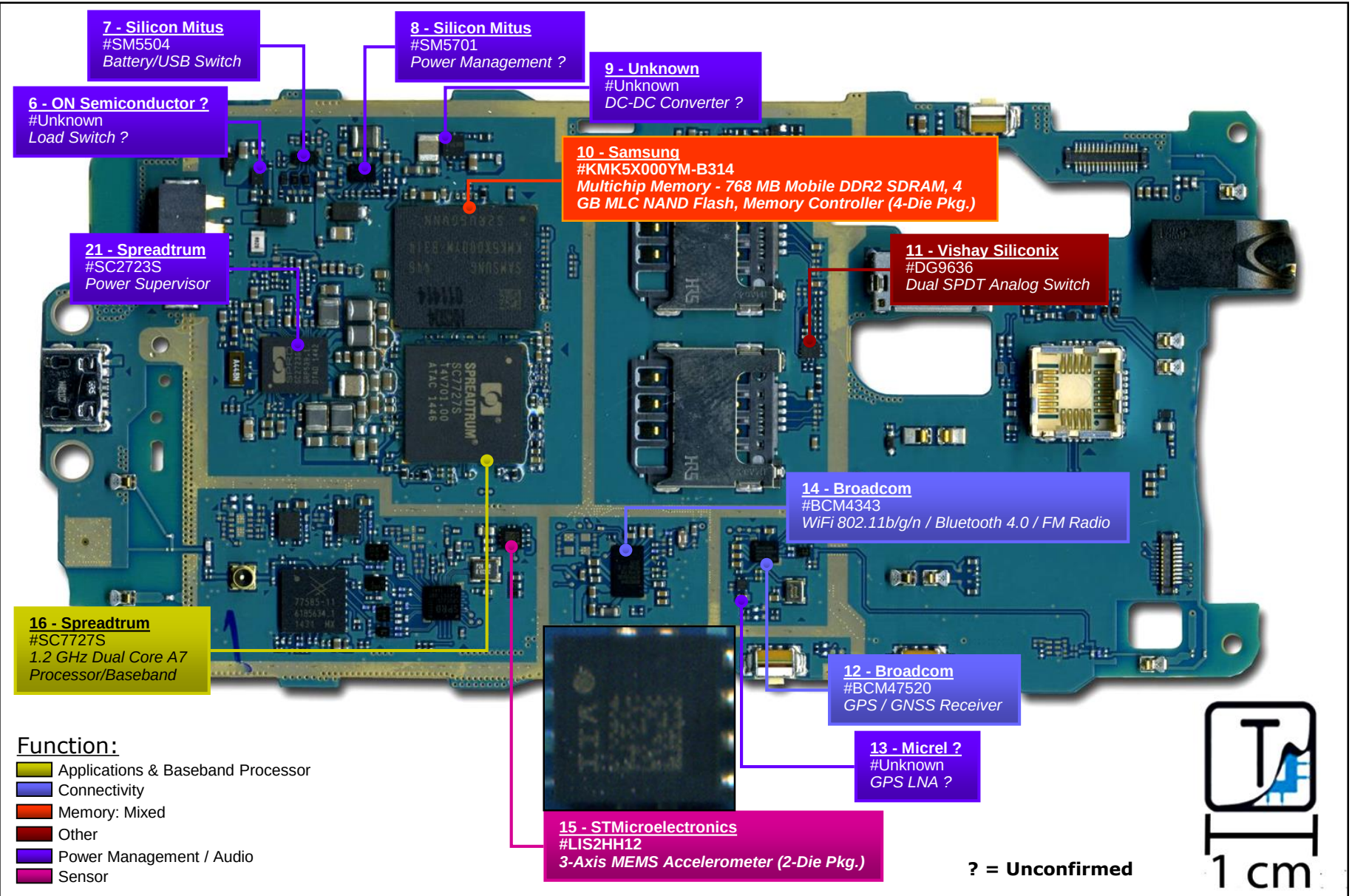
These categories are in alignment with the high-level IC function categories currently found in IRIS 2.0. The categories will also be found in the updated Excel BoM workbooks included with each final report.

Component Function	
Applications/Baseband Processor	Processors that are both Applications/Baseband Processors
Application Processor	Main Processor without Cellular Modem
Baseband Processor	Cellular Modem Processor
Camera / Image	Image Sensors, Video processors, Image Coprocessors, VGA Camera
Connectivity	WiFi, Bluetooth, GPS, USB, ZigBee Components
Display / Touchscreen	Display Driver, Touchscreen Controller, etc.
Memory: Mixed	Components with both non-volatile and volatile memory
Memory: Non-Volatile	NAND, NOR, EEPROMs, etc.
Memory: Volatile	RAM, SDRAM, etc.
Mixed Signal	DSPs, MCUs, Mixed-Signal Arrays, Processors (non-audio, non-apps, non-baseband)
Other	Small Logic AND, OR Gates, LDOs, Transistors, Regulators
Power Management / Audio	Audio CODECs, Envelope Tracking, Power Management
RF Component	RF Antenna Switches, RF Filters, RF Power Amplifiers, RF Receivers, RF Transceivers, etc.
Sensor	Accelerometer, Gyroscope, Barometer, Heart Rate, MEMS Microphones, Pressure, Temperature, Ambient Light/Proximity

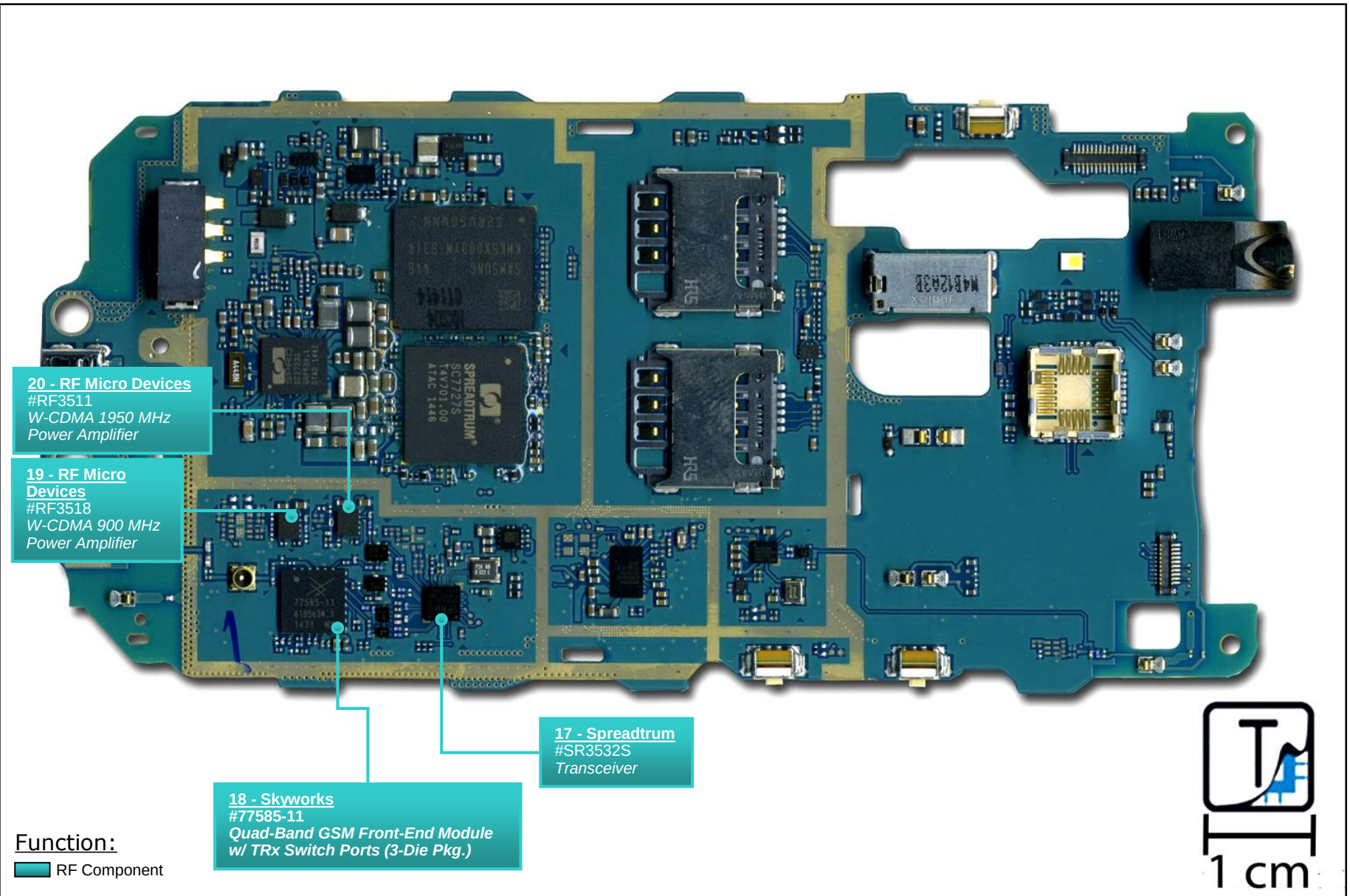
Main Board (Side 1)



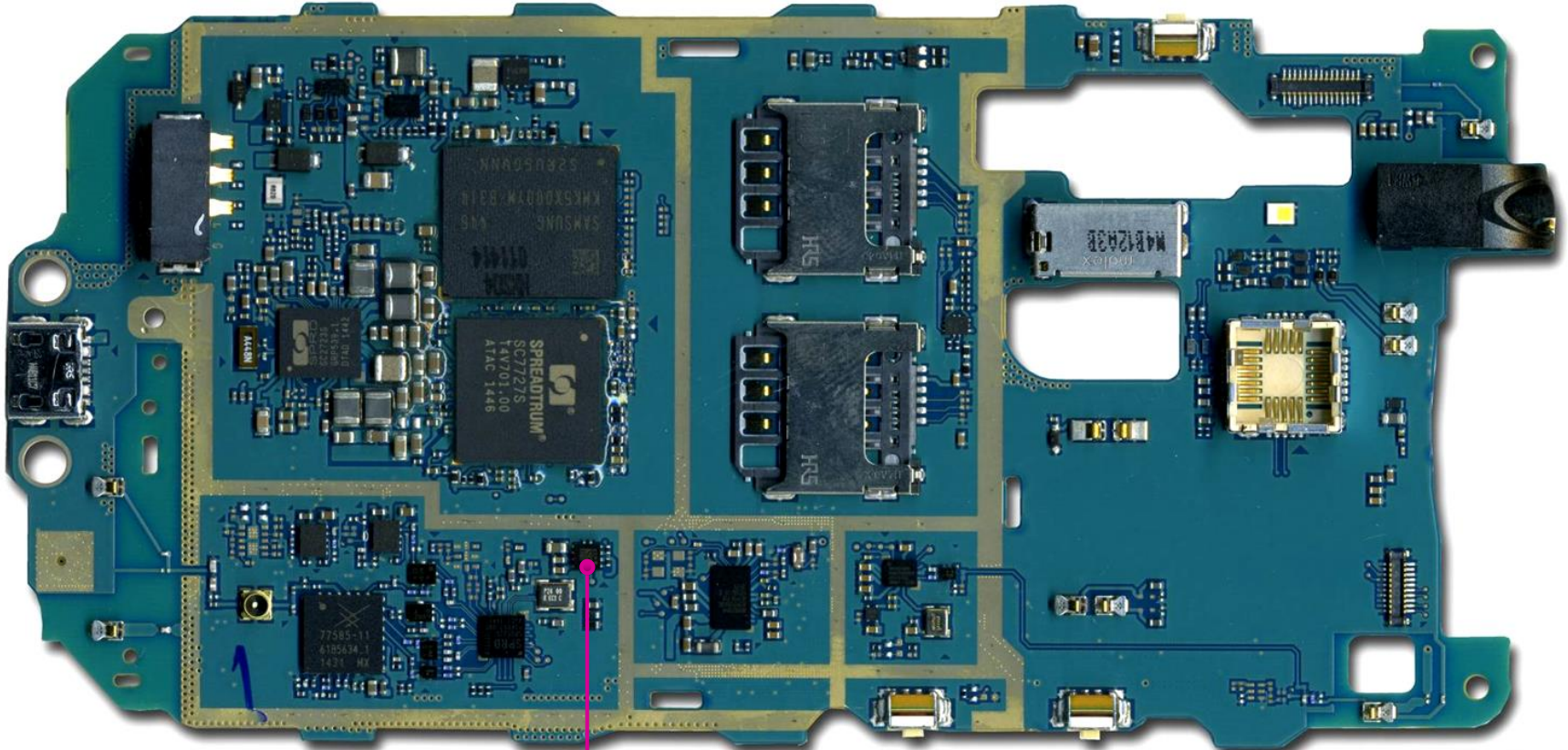
Main Board (Side 1 ICs)



Main Board (Side 1 ICs - Radio)



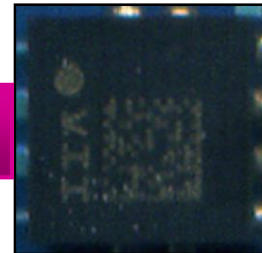
Main Board (Side 1 ICs - Sensors)



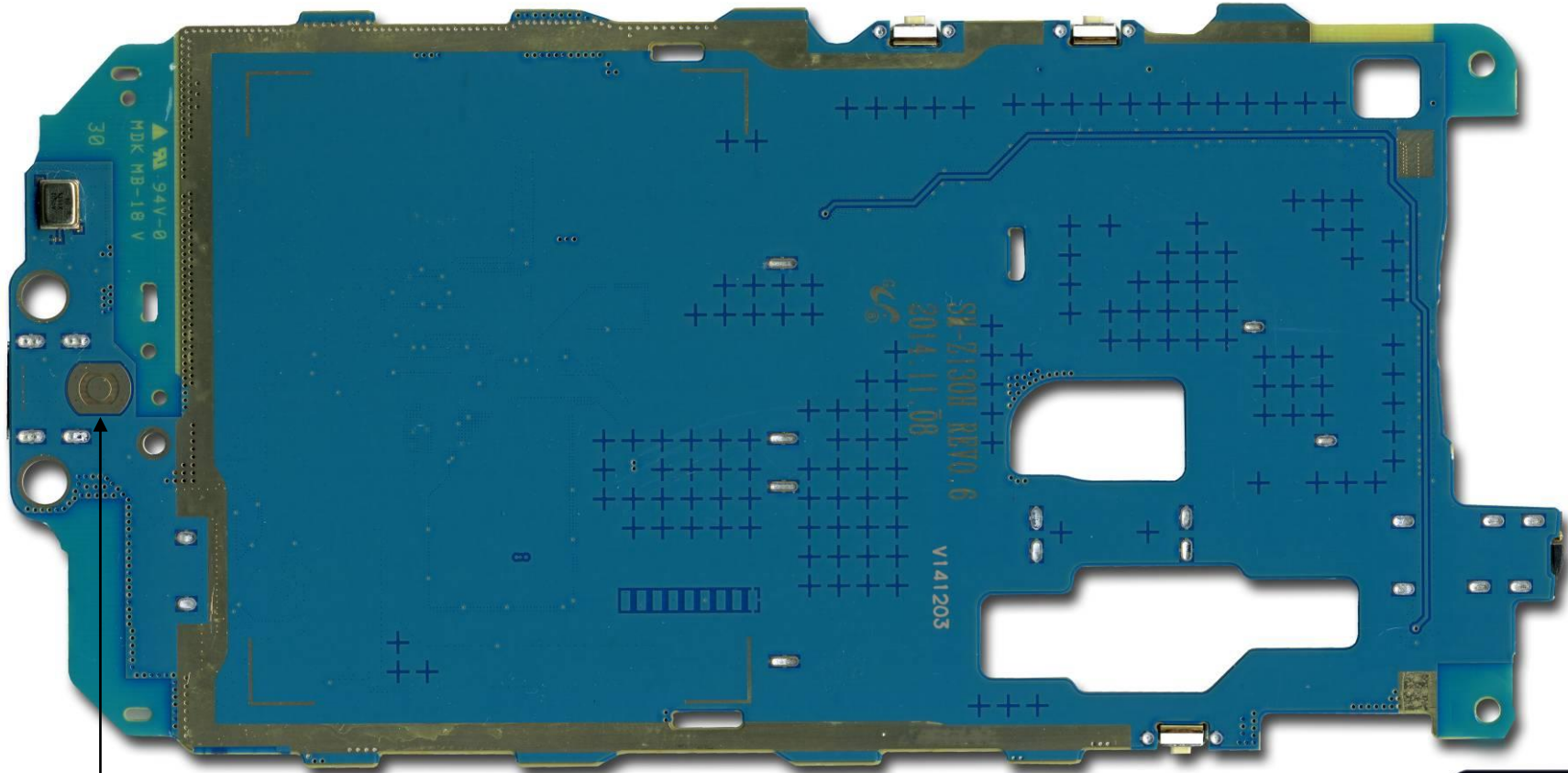
Function:

Sensor

15 - STMicroelectronics
#LIS2HH12
3-Axis MEMS Accelerometer (2-Die Pkg.)



Main Board (Side 2)

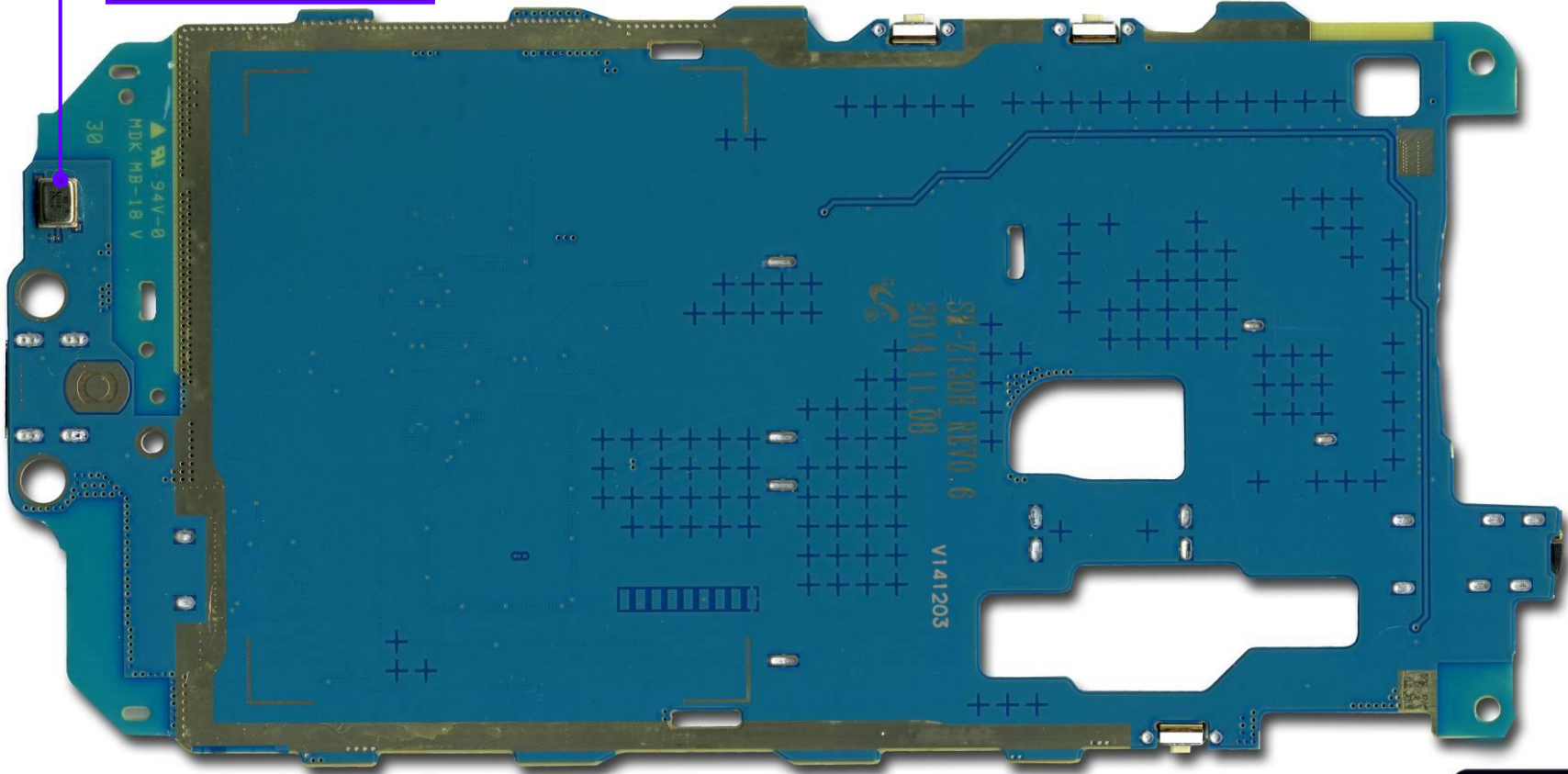


Home Key Button



Main Board (Side 2 ICs)

22 - Cirrus Logic ?
#WM7132PE ?
MEMs Microphone (2-Die Pkg.)



Function:

Power Management / Audio

? = Unconfirmed

